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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC A) – MG5DSCBTG301- ENZYME TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word answer-Answer any 10 questions.
Each question carries 2 marks

1. 1. Which enzyme helps digest starch? [K] / [CO1]
   Amylase- 2 marks.
2. What are biological catalysts called? [K] / [CO1]
   Enzymes- 2 marks.
3. What happens to an enzyme's structure when it loses its functional conformation? [U] / [CO1]
   Denaturation (resulting in reduced or complete loss of catalytic activity.)- 2 marks
4. Recall the EC class involved in molecular rearrangements. [K] / [CO1]
   Isomerases (EC Class 5) – 2 marks.
5. What type of inhibitor competes with the substrate for the active site? [K] / [CO2]
   Competitive inhibitor- 2 marks.
6. Connect enzyme stability with enzyme shelf life. [An]/ [CO4]
   Enzyme stability increases shelf life- 2 marks.
7. Name the two techniques in which an enzyme is respectively attached to a surface and trapped inside a gel. [K] / [CO4]
   Adsorption, Entrapment- 2 marks.
8. Explain the role of GGT in the interpretation of increased ALP. [U] / [CO4]
   Elevated GGT along with ALP suggests a hepatobiliary source of increased ALP, whereas normal GGT may suggest a non-hepatic source or hepatobiliary disease or (liver disease)- 2 marks
9. Name two major sources of industrial enzymes. [K] / [CO4]
   Microorganisms and plants/animals or fungi are two major sources of industrial enzymes.- 2 marks
10. What are PETases? [U] / [CO4]

PETases are enzymes that degrade polyethylene terephthalate (PET) by hydrolysing its ester bonds/PETases are enzymes capable of degrading polyethylene terephthalate (PET) into smaller components.- 2 marks

11. Analyse the relationship between enzyme specificity and the efficiency of enzymatic bioremediation. [An] / [CO4]

    Enzyme specificity enables selective degradation of target pollutants, improving bioremediation efficiency and reducing unwanted reactions-2 marks.

12. Which phenomenon causes reduced reaction rates inside immobilized enzyme matrices? [U] / [CO4]

    Diffusion limitation-2 marks

Part B

Short essay-Answer any 4 questions.

Each question carries 5 marks

13. Compare absolute specificity, group specificity, and stereo- specificity with suitable examples. [U] / [CO1]

    Enzyme specificity- introduction-.5 marks, Absolute specificity-definition- refers to an enzyme acting on one particular substrate , example- urease–urea etc – 1 .5 marks, group specificity – definition-involves substrates with particular structural features, example-trypsin–peptide bonds after lysine or arginine etc, - 1.5 marks, stereo-specificity- definition- involves recognition of a particular stereoisomer , example- L-amino acid oxidase–L-amino acids etc- 1.5 marks.

14. Classify fumarase based on its catalytic activity and explain the type of reaction it performs. [U] / [CO1]

    Lyase class - introduction- 1 mark, explanation- Fumarase is a lyase (EC 4; EC 4.2.1.2) that catalyses the reversible hydration of fumarate to L-malate and dehydration of L-malate to fumarate in the citric acid cycle. Any 4 points- 4 marks.

15. Connect improved catalytic activity of engineered enzymes with increased industrial productivity. [An] / [CO4]

    Engineered enzyme – Definition- 1 mark, explanation-Engineered enzymes with improved catalytic activity increase reaction rates and product yield, while enhanced stability reduces enzyme replacement and processing costs, thereby increasing industrial productivity. (Any 4 points)- 4 marks.

16. Describe how binding of a substrate at one active site can affect the other active sites of an allosteric enzyme. [U] / [CO3].

    Allostric enzyme – Definition-1 mark, explanation-Substrate binding at one active site induces a conformational change that affects the other active sites; positive cooperativity increases their substrate affinity, whereas negative cooperativity decreases it. (Any 4 points)- 4 marks.

17. Describe the effect of temperature on enzyme activity. [U] / [CO2]

    Factors affecting enzyme activity-introduction- 1 mark, explanation-Enzyme activity generally increases with temperature up to an optimum due to increased molecular movement, but above the optimum, denaturation causes a decrease in activity- 3 marks, Graphical Representation -1 mark.

18. Explain the role of synthetic enzymes in biosensors. [U] / [CO4]

Synthetic enzyme – Definition- 1 mark, explanation-act as catalytic recognition components that generate measurable signals in response to target analytes, improving biosensor sensitivity, stability and diagnostic applications. Any 4 points)- 4 marks

Part C

Essay Questions-Answer any 1 question.

Each question carries 10 marks

19. Explain cofactors and their role in enzyme catalysis. [U] / [CO1]

    Cofactors - Introduction -2 marks, Types of cofactor – organic, inorganic and prosthetic groups with examples - 4 marks. Role of cofactor -4marks.

20. Explain the Michaelis–Menten equation in detail and discuss the significance of Km and Vmax in enzyme kinetics.[U] / [CO2]

    The Michaelis–Menten equation -Introduction -2 marks, Derivation of the equation-6 marks, significance -2 marks.